

LEC 4

Hybrid Systems

Introduction

- Cyber-physical systems integrate physical dynamics and computational systems, so they commonly combine both discrete and continuous dynamics.
- In this chapter, we show that the modeling techniques of Chapters 2 and 3 can be combined, yielding what are known as hybrid systems.
- Hybrid system models are often much simpler and more understandable.
- They are a powerful tool for understanding real-world systems.

Modal Models

- In this section, we show that state machines can be generalized to admit continuous inputs and outputs and to combine discrete and continuous dynamics.
- A hybrid system is sometimes called a modal model because it has a finite number of modes, one for each state of the FSM.
- The states of the FSM may be referred to as modes rather than states

Actor Model for State Machines

- It is natural to think of a state machine as an actor, as illustrated in Figure

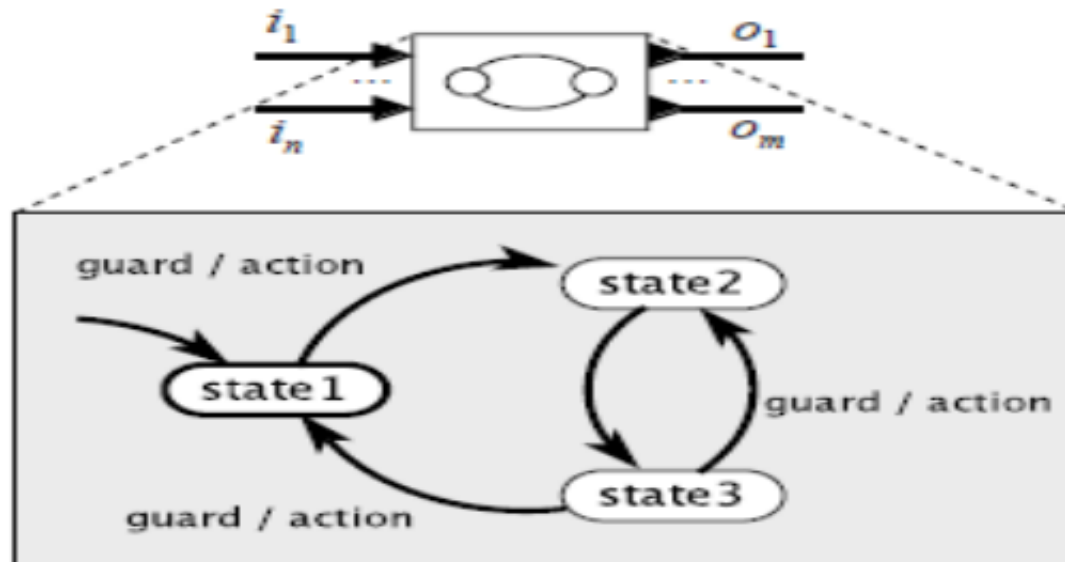


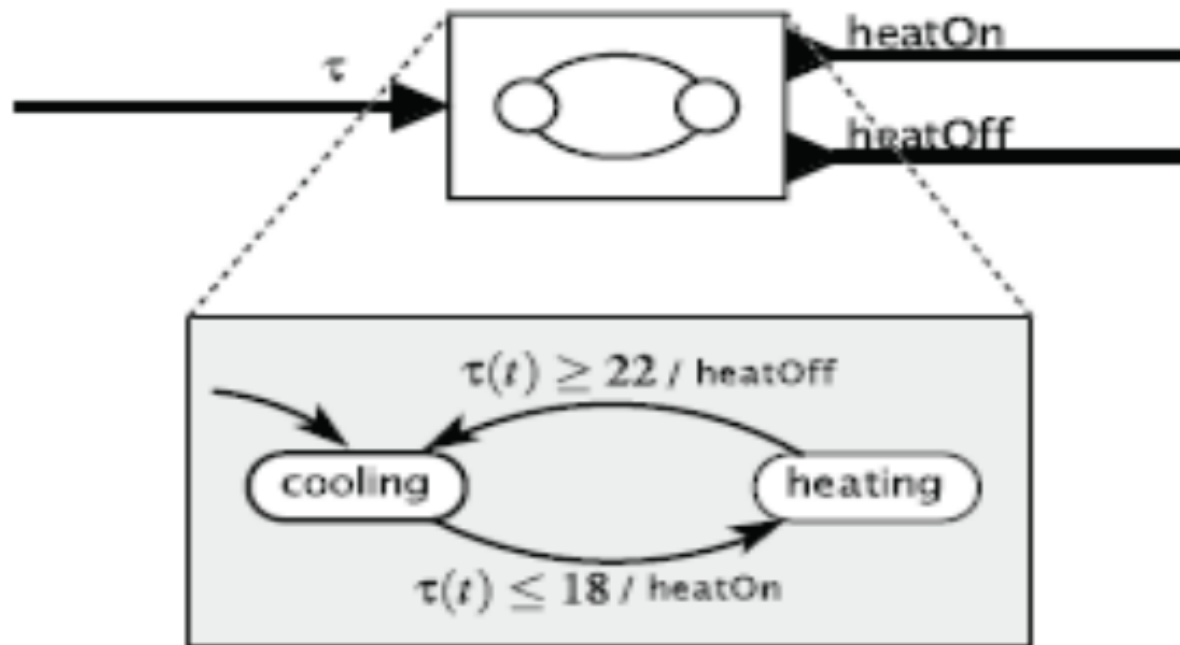
Figure 4.1: An FSM represented as an actor.

What About Continuous Inputs?

- We have so far assumed that state machines operate in a sequence of discrete reactions.
- We have assumed that inputs and outputs are absent between reactions.
- We will now generalize this to allow inputs and outputs to be continuous-time signals.
- In order to get state machine models to coexist with time-based models, we need to interpret state transitions to occur on the same timeline used for the time-based portion of the system.

Example 1

- Consider a thermostat modeled as a state machine with states = {heating; cooling}, shown in Figure.



Cont.,

- The input is a continuous-time signal $T : \mathbb{R} \rightarrow \mathbb{R}$ where $T(t)$ represents the temperature at time t .
- The initial state is cooling, and the transition out of this state is enabled at the earliest time t after the start time when $T(t) \leq 18$.
- In this example, we assume the outputs are pure signals {heatOn and heatOff}.

What About Continuous Outputs?

- In the above example, the outputs are present only at the times the transitions are taken.
- We can also generalize FSMs to support continuous-time outputs, but to do this, we need the notion of state refinements.
- A hybrid system associates with each state of an FSM a dynamic behavior, called “**state refinements**” that gives the dynamic behavior of the output as a function of the input and (possibly) additional continuous state variables.

Example 2

- Suppose that instead of discrete outputs as in Example 1 we wish to produce a control signal whose value is 1 when the heat is on and 0 when the heat is off.
- Such a control signal could directly drive a heater.
- Each state has a refinement that gives the value of the output **h** while the state machine is in that state.

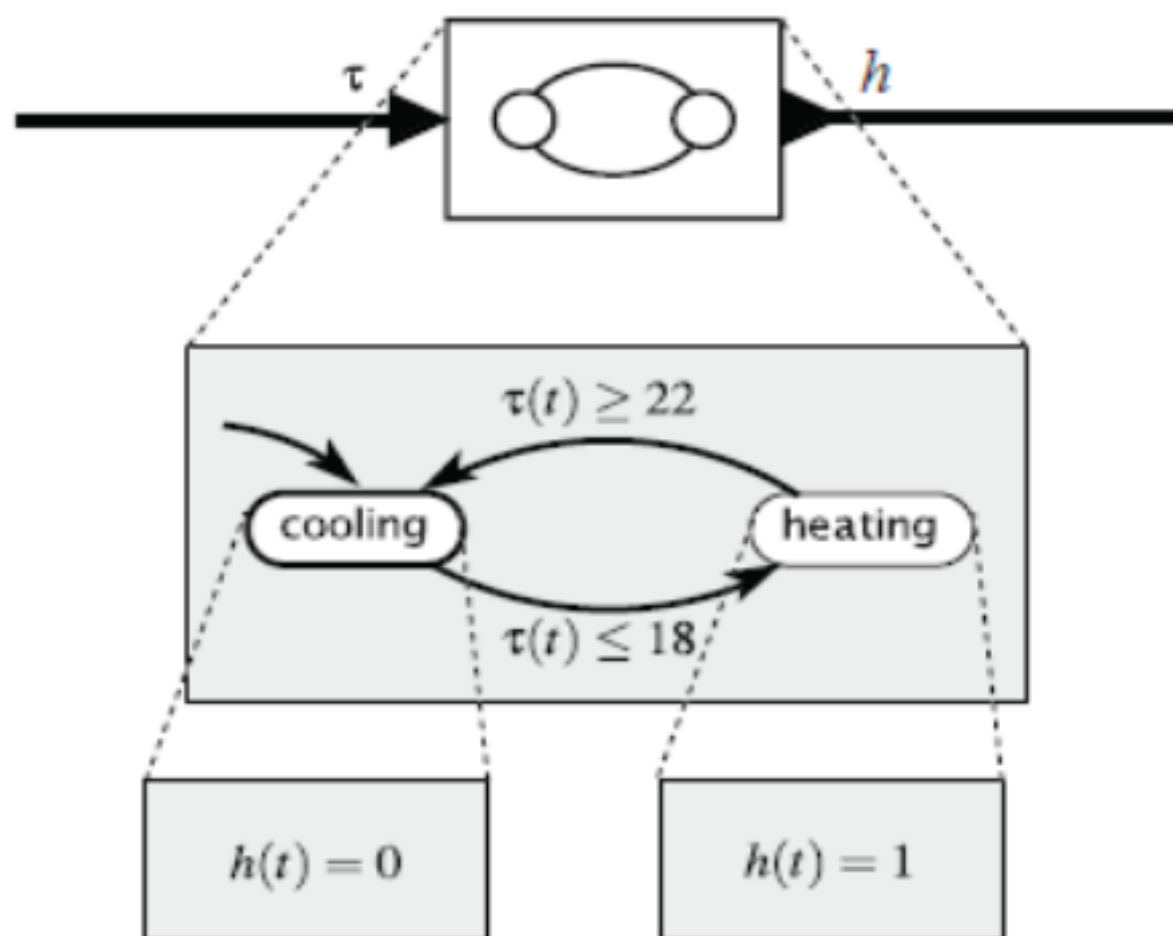


Figure 4.3: A thermostat with continuous-time output.

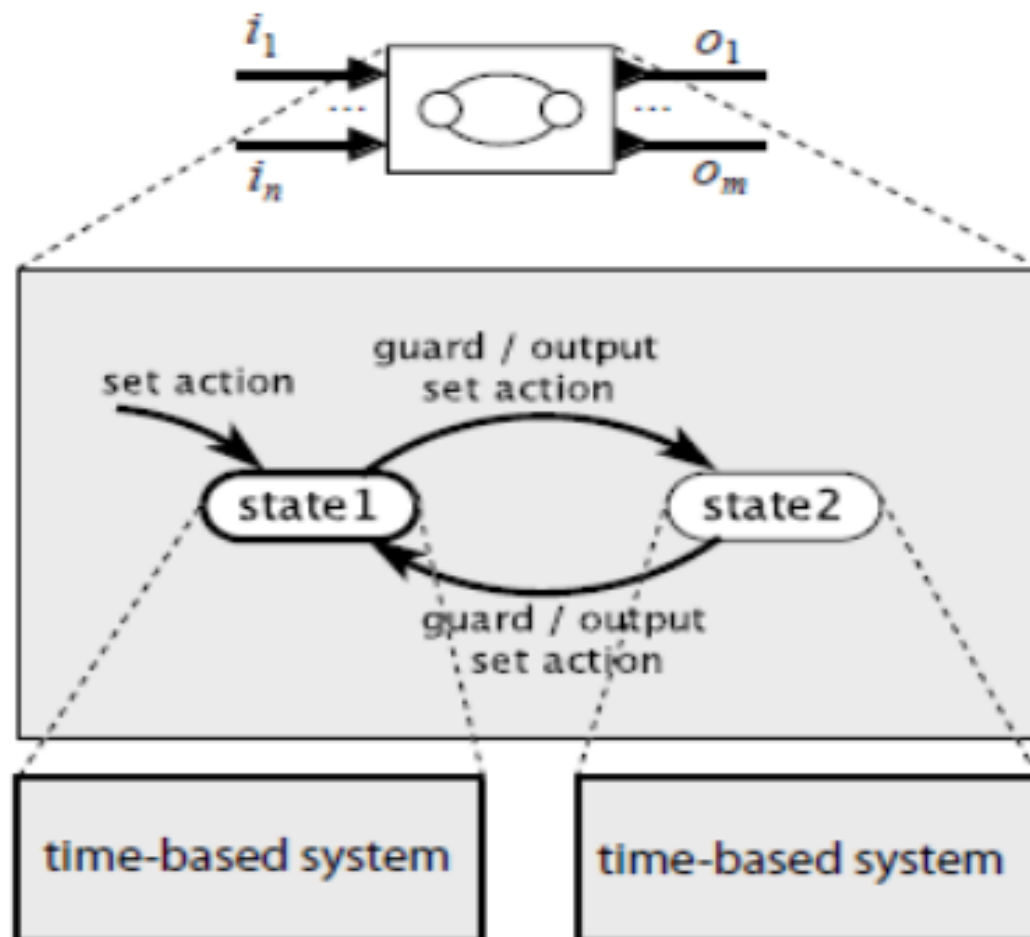


Figure 4.4: Notation for hybrid systems.

Classes of Hybrid Systems:

1- Timed Automata

- Most cyber-physical systems require measuring the passage of time and performing actions at specific times.
- A device that measures the passage of time, a clock, has a particularly simple dynamics: its state progresses linearly in time.
- Timed automata are the simplest non-trivial hybrid systems.
- They are modal models where the time-based refinements have very simple dynamics; all they do is measure the passage of time.

Clock Model

- A clock is modeled by a first-order differential equation,

$$\forall t \in T_m, \quad \dot{s}(t) = a,$$

where $s: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous-time signal, $s(t)$ is the value of the clock at time t , and $T_m \subset \mathbb{R}$ is the subset of time during which the hybrid system is in mode m . The rate of the clock, a , is a constant while the system is in this mode.¹

Example 3

- Recall the thermostat of Example 1, which uses hysteresis to prevent chattering.
- An alternative implementation that would also prevent chattering would use a single temperature threshold, but instead would require that the heater remain on or off for at least a minimum amount of time, regardless of the temperature.
- This can be modeled as a timed automaton.

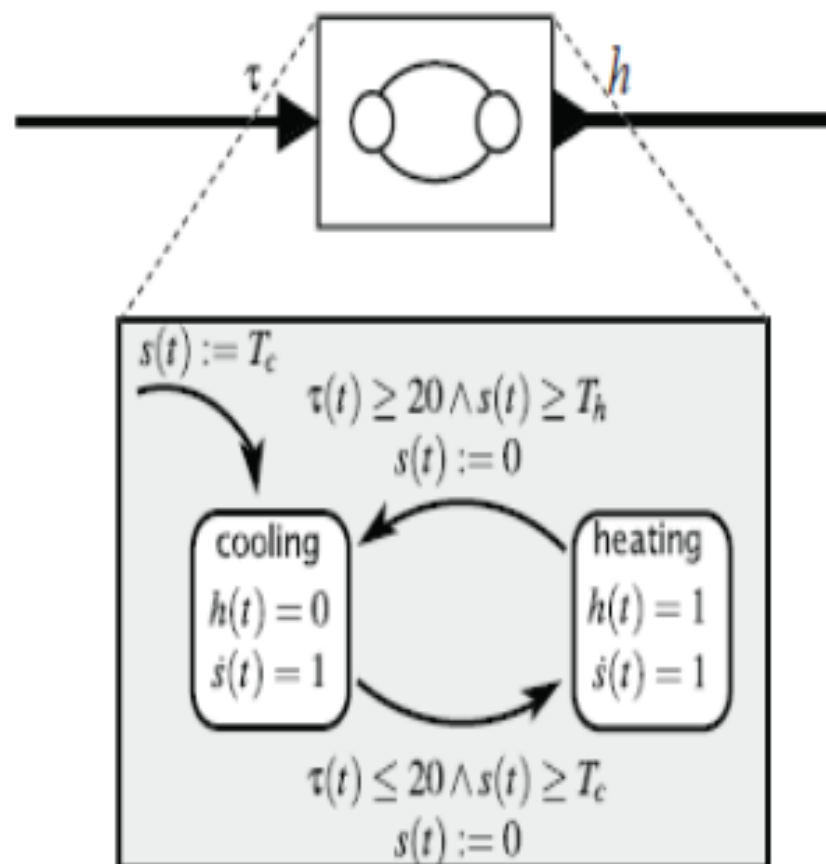
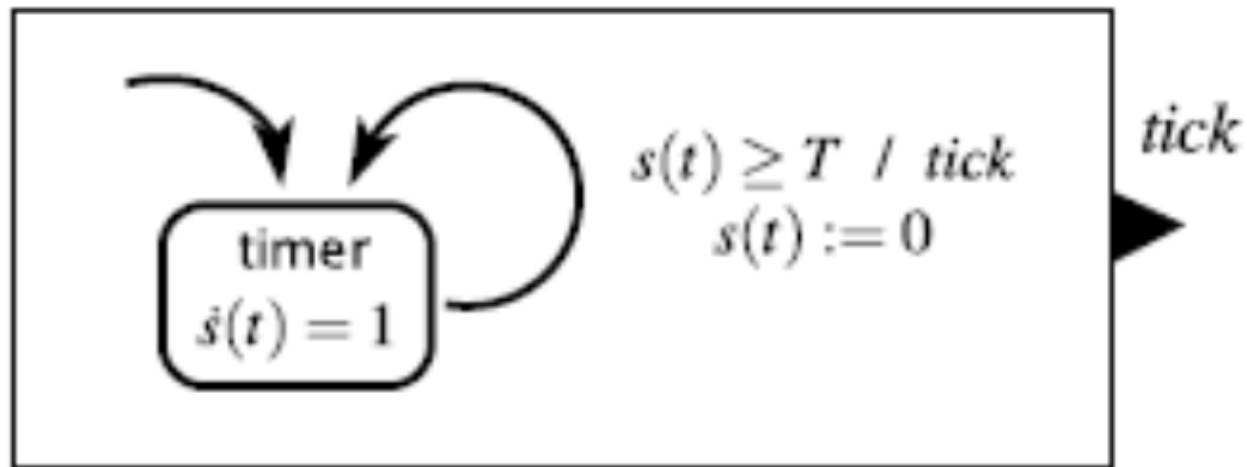


Figure 4.5: A timed automaton modeling a thermostat with a single temperature threshold, 20, and minimum times T_c and T_h in each mode.

Example 4

- A timed automaton produces a pure output that will be present every T time units, starting at the time when the system begins executing.



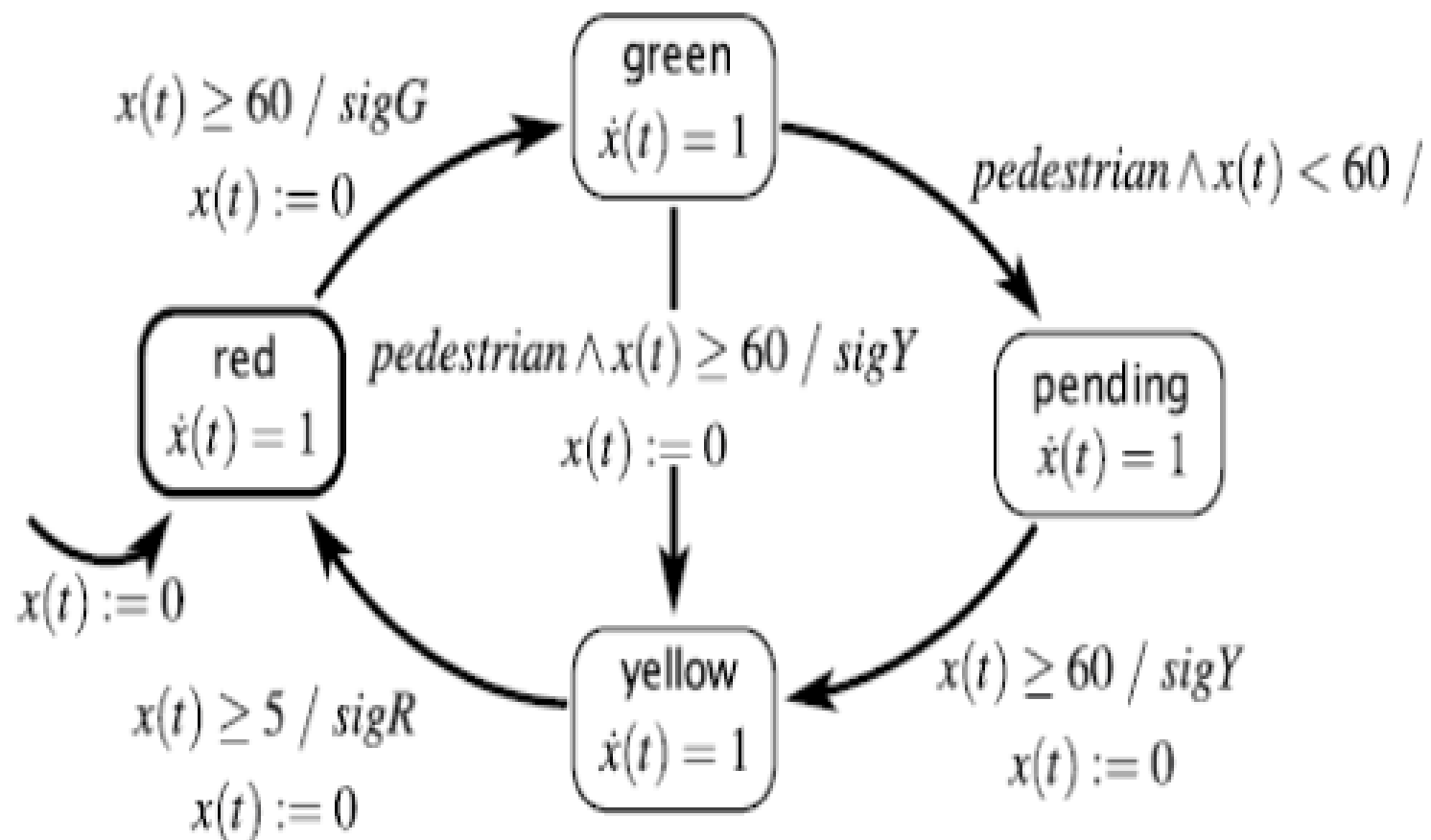
Example 5

- The traffic light controller is a time triggered machine that assumes it reacts once each second.

continuous variable: $x(t) : \mathbb{R}$

inputs: $pedestrian$: pure

outputs: $sigR, sigG, sigY$: pure



Sheet 4

- **Exercises:** 2, 4, 6, and 9.